

JOHNNY CHAN

also known as Zhe Chen

ROBOTICS SOLUTIONS ENGINEER | Robotics, Automation & Technical Project Coordination

Kuala Lumpur, Malaysia | +60 16 641 7042 | Johnny8426@outlook.com | jc8426.github.io | github.com/JC8426

PROFESSIONAL SUMMARY

Robotics and automation engineer with mechanical-engineering foundations and current doctoral research in computer science, UAV autonomy, and distributed robotic systems. Skilled in translating broad technical and operational goals into structured requirements, reviewable solution materials, and coordinated engineering tasks. Brings practical exposure to manufacturing workflows, supplier communication, production records, logistics, and cross-border technical collaboration in Chinese and English.

CORE COMPETENCIES

Technical Solutions: Requirement Analysis · Solution Definition · System-Level Engineering · Technical Proposal & Presentation Support

Project Delivery: Cross-functional Coordination · Supplier Communication · Project Documentation · Technical Handover · Delivery Tracking

Robotics & Automation: ROS Noetic · Gazebo · RViz · UAV Simulation · Path Planning · System Integration · Python · C++ · Linux

Industrial Exposure & Tools: Manufacturing Workflow · Production & Quality Records · Logistics Coordination · Git · Markdown · Excel

PROFESSIONAL EXPERIENCE

Research Engineer / PhD Candidate & Lab Coordinator | Robotedge Laboratory, Universiti Malaya — **2026–Present**
Malaysia

- Coordinate robotics research and laboratory workflows by translating broad goals into structured tasks, validation records, technical documentation, and reviewable handoff materials.
- Develop and review ROS/Gazebo workflows for UAV autonomy, simulation, path planning, and system-integration studies across software and laboratory environments.
- Support communication among students, researchers, and supervisors while organizing project records, experiment tracking, and technical resources.

Robotics Systems Research — Independent / Visiting Research Track | Malaysia / China **2023–2025**

- Developed UAV simulation environments for formation control, obstacle avoidance, and path-planning studies using ROS Noetic, Gazebo, RViz, Python, and C++.
- Reproduced and analyzed EGO-Planner and EGO-Swarm workflows to understand interactions among sensing, planning, control, software, and operating environments.
- Documented configurations, validation findings, and reproducible workflows to support technical review and knowledge transfer.

R&D Project Engineer / Technical Coordinator | Suzhou Xili Textile / Suzhou Sushanjin Biotechnology — **2021–2025**
China

- Coordinated R&D execution across technical, design, supplier, production, and management stakeholders, supporting work from preparation through finished-product delivery.
- Supported raw-material selection, supplier communication, production planning, manufacturing-process review, and finished-product logistics and transportation.
- Prepared production and quality records, exhibition and customer-facing materials, and structured project documentation for internal review and external communication.

SELECTED PROJECTS

UAV Simulation & Robotics Integration Platform — Integrated ROS Noetic, Gazebo, RViz, Python, C++, and Linux workflows for autonomy, path-planning, obstacle-avoidance, and multi-robot studies.

Industrial Vision Route Evaluation — Supported supplier comparison, automation-route assessment, integration planning, and technical-commercial option review.

AI-assisted Engineering & Documentation Workflow — Structured coding support, experiment tracking, technical documentation, and project handoff using Codex, ChatGPT, Claude, Git, and Markdown.

EDUCATION

PhD in Computer Science, Universiti Malaya, Department of AI, FCSIT | 2026–Present

M.Eng. in Mechanical Engineering, Xinjiang University, China | 2016–2019

B.Eng. in Mechanical Engineering, Xinjiang University, China | 2012–2016

LANGUAGES

Chinese — Native | English — Advanced, IELTS 7.0 / C1 | Bahasa Melayu — Beginner